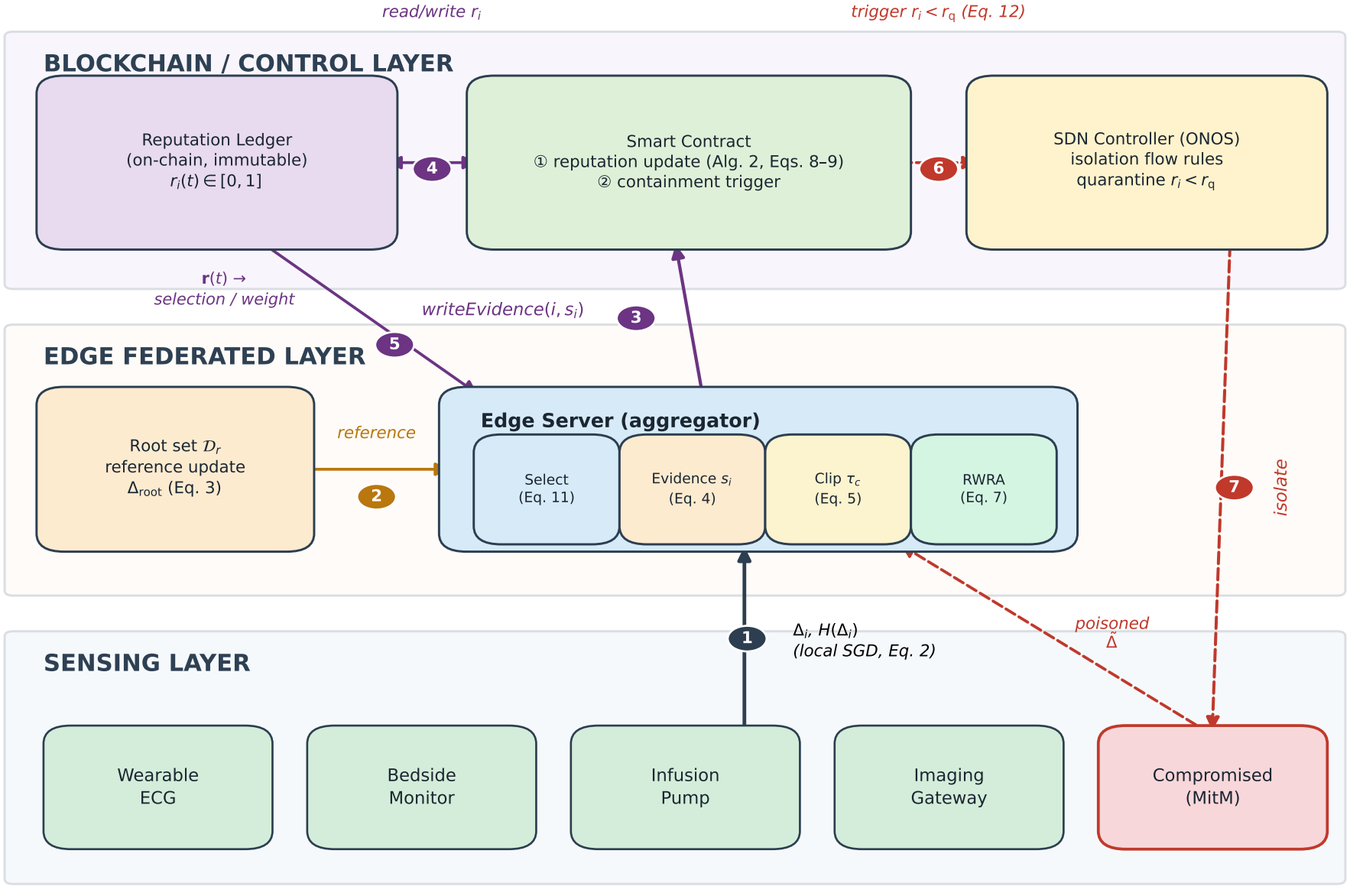
read/write r_i
trigger $r_i < r_q$ (Eq. 12)
BLOCKCHAIN / CONTROL LAYER
Reputation Ledger
(on-chain, immutable)
$r_i(t) \in [0, 1]$
4
Smart Contract
① reputation update (Alg. 2, Eqs. 8–9)
② containment trigger
6
SDN Controller (ONOS)
isolation flow rules
quarantine $r_i < r_q$
$\mathbf{r}(t) \rightarrow$
selection / weight
writeEvidence(i, s_i)
3
5
EDGE FEDERATED LAYER
Root set $\mathcal{D}_r$
reference update
Δ_{root} (Eq. 3)
reference
2
Edge Server (aggregator)
Select
(Eq. 11)
Evidence s_i
(Eq. 4)
Clip τ_c
(Eq. 5)
RWRA
(Eq. 7)
7
isolate
1
Δ_i, $H(\Delta_i)$
(local SGD, Eq. 2)
poisoned
$\tilde{\Delta}$
SENSING LAYER
Wearable
ECG
Bedside
Monitor
Infusion
Pump
Imaging
Gateway
Compromised
(MitM)
Legend:
training / data flow
on-chain reputation
control / isolation
root reference